

## **BUSINESS INTELLIGENCE AND TOOLS**

### **EXPERIMENT - 4**

#### **Problem File**

**Title:** OLAP Operations using Microsoft Power BI

**Aim:** To perform OLAP operations such as Slice, Dice, Drill-down, and Drill-up on the Global Superstore dataset using Microsoft Power BI.

#### **Objectives**

- To understand the concept of OLAP in Business Intelligence.
- To load and analyze the Global Superstore dataset using Power BI.
- To create visualizations for sales and profit analysis.
- To perform Slice operations using selected dimensions.
- To perform Dice operations using multiple filtering conditions.
- To perform Drill-down analysis from higher to lower levels of detail.
- To perform Drill-up analysis from lower to higher levels of detail.
- To interpret the results obtained from multidimensional analysis.

#### **Tools Required**

- Microsoft Power BI Desktop
- Microsoft Excel
- Power Query
- Global Superstore 2018 dataset

**Dataset Description:** The Global Superstore 2018 Excel dataset contains three sheets:

**Orders:** Contains 51,290 records and 24 columns related to orders and sales.

Important attributes include:

- Row ID
- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- City
- State
- Country
- Region
- Market
- Product ID
- Product Name
- Sub-Category
- Category
- Sales
- Quantity
- Discount
- Profit
- Shipping Cost
- Order Priority

**Returns:** Contains 2,033 records related to returned orders.

Important fields include:

- Returned
- Order ID
- Region

**People:** Contains 24 records mapping people to regions.

Important fields include:

- Person
- Region

## Expected Output

After completing the experiment, the following outputs should be obtained:

- Sales by Region visualization.
- Profit by Category visualization.
- Sales by Segment visualization.
- Year-wise Sales Trend.
- Sliced data for Western Europe.
- Diced data for Western Europe, Technology, and 2017.
- Region → State → City hierarchy.
- Successful Drill-down and Drill-up operations.
- Observations explaining the changes at different levels of analysis.